

FL Simulation Results

File: test_A3

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 70.0% (0.70)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 70.0% (0.70) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8735
Clean + DP Protection Accuracy: 0.8723
Poisoned Accuracy: 0.8094
Poisoned + DP Protection Accuracy: 0.8084
Drop (Clean - Poisoned): 0.0641
Drop (Clean - Init): 0.0807
Drop (Clean DP - Init): 0.0795
Drop (Poisoned - Init): 0.0166
Drop (Poisoned DP - Init): 0.0156
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8800

Clean Recall: 0.8735

Clean F1 Score: 0.8729

Clean DP Precision: 0.8792

Clean DP Recall: 0.8723

Clean DP F1 Score: 0.8717

Poisoned Precision: 0.8202

Poisoned Recall: 0.8094

Poisoned F1 Score: 0.8087

DP Protection Precision: 0.8196

DP Protection Recall: 0.8084

DP Protection F1 Score: 0.8076

Summary

FL Simulation Complete

Task: 86c27c44-e494-41a9-b3cb-917edb2d2126

GPU: GPU 0

Init Accuracy: 0.7928

Clean Accuracy: 0.8735

Clean DP Accuracy: 0.8723

Poisoned Accuracy: 0.8094

Data Poison Protection Accuracy: 0.8084

Drop (Clean - Poisoned): 0.0641

Drop (Clean - Init): 0.0807

Drop (Clean DP - Init): 0.0795

Drop (Poisoned - Init): 0.0166

Drop (Poisoned_DP - Init): 0.0156

Data Poison Protection Method: foolsgold

--- Confusion Matrix Metrics (Weighted Avg) ---

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DP Protection Precision: 0.8196

DP Protection Recall: 0.8084

DP Protection F1 Score: 0.8076